

Steps to install Hadoop:

1. Make sure java is installed.

```
java -version
```

If java is not installed, then type in the following commands:

```
sudo apt-get install update
```

```
sudo apt-get update
```

```
sudo apt-get install default-jdk
```

Make sure now java is installed.

```
java -version
```

```
husseinfadl@husseinfadl:/home$ java -version
openjdk version "11.0.10" 2021-01-19
OpenJDK Runtime Environment (build 11.0.10+9-Ubuntu-0ubuntu1.18.04)
OpenJDK 64-Bit Server VM (build 11.0.10+9-Ubuntu-0ubuntu1.18.04, mixed mode, sharing)
husseinfadl@husseinfadl:/home$
```

2. Install ssh server

```
sudo apt-get install ssh-server
```

Generate public/private RSA key pair.

```
ssh-keygen -t rsa -P ""
```

When prompted for the file name to save the key, press Enter (leave it blank).

```
husseinfadl@husseinfadl:/home$ ssh-keygen -t rsa -P ""
Generating public/private rsa key pair.
Enter file in which to save the key (/home/husseinfadl/.ssh/id_rsa):
Your identification has been saved in /home/husseinfadl/.ssh/id_rsa.
Your public key has been saved in /home/husseinfadl/.ssh/id_rsa.pub.
The key fingerprint is:
SHA256:XqVfPkZlmsIG7z9I1pX/zMJZo2hWX7K9HtcgjIjLEyQ husseinfadl@husseinfadl
The key's randomart image is:
+---[RSA 2048]---+
|
|      E . . . + |
|      o . . 0 * . |
|      o S + B.*.. |
|      . + . +oBoo= |
|      + . o=o*X* |
|      .   +.*oB |
|      o   o=. |
+---[SHA256]-----+
```

Type the following commands:

```
cat $HOME/.ssh/id_rsa.pub >> $HOME/.ssh/authorized_keys
```

```
ssh localhost
```

```
exit
```

```
husseinfadl@husseinfadl:~$ ssh localhost
Welcome to Ubuntu 18.04.5 LTS (GNU/Linux 4.15.0-139-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

 * Canonical Livepatch is available for installation.
   - Reduce system reboots and improve kernel security. Activate at:
     https://ubuntu.com/livepatch

80 packages can be updated.
27 of these updates are security updates.
To see these additional updates run: apt list --upgradable

New release '20.04.2 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Tue Apr 13 15:29:36 2021 from 127.0.0.1
husseinfadl@husseinfadl:~$
```

```
husseinfadl@husseinfadl:~$ exit
logout
Connection to localhost closed.
husseinfadl@husseinfadl:~/home$
```

3. Install Hadoop by navigating to the following link and downloading the tar.gz file for Hadoop version 3.3.0 (or a later version if you wish). (478 MB)

<https://hadoop.apache.org/release/3.3.0.html>

Release 3.3.0 available

This is the first release of Apache Hadoop 3.3 line. It contains 2148 bug fixes, improvements and enhancements since 3.2.

Users are encouraged to read the [overview of major changes](#). For details of please check [release notes](#) and [changelog](#).

2020 Jul 14

[Download tar.gz](#)[\(checksum signature\)](#)[Download aarch64 tar.gz](#)[\(checksum signature\)](#)[Download src](#)[\(checksum signature\)](#)[Documentation](#)

4. Once downloaded, open the terminal and cd to the directory where it is downloaded (assume the desktop for example) and extract it as follows:

cd Desktop

sudo tar -xvzf hadoop-3.3.0.tar.gz

You can now check that there is an extracted file named hadoop-3.3.0 by typing the command “ls” or by visually inspecting the files.

- Now, we move the extracted file to the location /usr/local/hadoop

sudo mv hadoop-3.3.0 /usr/local/hadoop

- Let's configure the hadoop system.

Type the following command:

sudo gedit ~/.bashrc

At the end of the file, add the following lines: (Note: Replace the java version with the version number you already have. You can navigate to the directory /usr/lib/jvm and check the file name java-xx-openjdk-amd64)

export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64

export HADOOP_HOME=/usr/local/hadoop

export PATH=\$PATH:\$HADOOP_HOME/bin

export PATH=\$PATH:\$HADOOP_HOME/sbin

export HADOOP_MAPRED_HOME=\$HADOOP_HOME

export HADOOP_COMMON_HOME=\$HADOOP_HOME

export HADOOP_HDFS_HOME=\$HADOOP_HOME

export YARN_HOME=\$HADOOP_HOME

export HADOOP_COMMON_LIB_NATIVE_DIR=\$HADOOP_HOME/native

export HADOOP_OPTS="-Djava.library.path=\$HADOOP_HOME/native"

```
# enable programmable completion features (you don't need to enable
# this, if it's already enabled in /etc/bash.bashrc and /etc/profile
# sources /etc/bash.bashrc).
if ! shopt -oq posix; then
  if [ -f /usr/share/bash-completion/bash_completion ]; then
    . /usr/share/bash-completion/bash_completion
  elif [ -f /etc/bash_completion ]; then
    . /etc/bash_completion
  fi
fi

export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
export HADOOP_HOME=/usr/local/hadoop
export PATH=$PATH:$HADOOP_HOME/bin
export PATH=$PATH:$HADOOP_HOME/sbin
export HADOOP_MAPRED_HOME=$HADOOP_HOME
export HADOOP_COMMON_HOME=$HADOOP_HOME
export HADOOP_HDFS_HOME=$HADOOP_HOME
export YARN_HOME=$HADOOP_HOME
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/native
export HADOOP_OPTS="-Djava.library.path=$HADOOP_HOME/native"
```

7. Save the file and close it.
8. Now from the terminal, type the following command:

```
source ~/.bashrc
```

9. We start configuring Hadoop by opening **hadoop-env.sh** as follows:

```
sudo gedit /usr/local/hadoop/etc/hadoop/hadoop-env.sh
```

Search for the line starting with **export JAVA_HOME=** and replace it with the following line.

```
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
```

Save the file by clicking on “Save” or (Ctrl+S)

```
# The java implementation to use. By default, this environment  
# variable is REQUIRED on ALL platforms except OS X!  
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
```

```
# Location of Hadoop. By default, Hadoop will attempt to determine  
# this location based upon its execution path.  
# export HADOOP_HOME=
```

```
# Location of Hadoop's configuration information. i.e., where this  
# file is living. If this is not defined, Hadoop will attempt to  
# locate it based upon its execution path.  
#
```

10. Open **core-site.xml** as follows:

```
sudo gedit /usr/local/hadoop/etc/hadoop/core-site.xml
```

Add the following lines between the tags <configuration> and </configuration> and save it (Ctrl+S).

```
<property>
  <name>fs.default.name</name>
  <value>hdfs://localhost:9000</value>
</property>

<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<!--
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  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
  <property>
    <name>fs.default.name</name>
    <value>hdfs://localhost:9000</value>
  </property>
</configuration>
```

11. Open **hdfs-site.xml** as follows:

```
sudo gedit /usr/local/hadoop/etc/hadoop/hdfs-site.xml
```

Add the following lines between the tags <configuration> and </configuration> and save it (Ctrl+S).

```
<property>
  <name>dfs.replication</name>
  <value>1</value>
</property>
<property>
  <name>dfs.namenode.name.dir</name>
  <value>file:/usr/local/hadoop_space/hdfs/namenode</value>
</property>
<property>
  <name>dfs.datanode.data.dir</name>
  <value>file:/usr/local/hadoop_space/hdfs/datanode</value>
</property>

<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<!--
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  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
  <property>
    <name>dfs.replication</name>
    <value>1</value>
  </property>
  <property>
    <name>dfs.namenode.name.dir</name>
    <value>file:/usr/local/hadoop_tmp/hdfs/namenode</value>
  </property>
  <property>
    <name>dfs.datanode.data.dir</name>
    <value>file:/usr/local/hadoop_tmp/hdfs/datanode</value>
  </property>
</configuration>
```

12. Open **yarn-site.xml** as follows:

```
sudo gedit /usr/local/hadoop/etc/hadoop/yarn-site.xml
```

Add the following lines between the tags <configuration> and </configuration> and save it (Ctrl+S)

```
<property>
  <name>yarn.nodemanager.aux-services</name>
  <value>mapreduce_shuffle</value>
</property>
<property>
  <name>yarn.nodemanager.aux-services.mapreduce.shuffle.class</name>
  <value>org.apache.hadoop.mapred.ShuffleHandler</value>
</property>
```

```
<?xml version="1.0"?>
```

```
<!--
```

```
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  you may not use this file except in compliance with the License.
  You may obtain a copy of the License at
```

```
    http://www.apache.org/licenses/LICENSE-2.0
```

```
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  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License. See accompanying LICENSE file.
```

```
-->
```

```
<configuration>
```

```
<!-- Site specific YARN configuration properties -->
```

```
<property>
```

```
  <name>yarn.nodemanager.aux-services</name>
```

```
  <value>mapreduce_shuffle</value>
```

```
</property>
```

```
<property>
```

```
  <name>yarn.nodemanager.aux-services.mapreduce.shuffle.class</name>
```

```
  <value>org.apache.hadoop.mapred.ShuffleHandler</value>
```

```
</property>
```

```
</configuration>
```

13. Open **mapred-site.xml** as follows:

```
sudo gedit /usr/local/hadoop/etc/hadoop/mapred-site.xml
```

Add the following lines between the tags <configuration> and </configuration> and save it (Ctrl+S)

```
<property>
  <name>mapreduce.framework.name</name>
  <value>yarn</value>
</property>
<property>
  <name>yarn.app.mapreduce.am.env</name>
  <value>HADOOP_MAPRED_HOME=${HADOOP_HOME}</value>
</property>
<property>
  <name>mapreduce.map.env</name>
  <value>HADOOP_MAPRED_HOME=${HADOOP_HOME}</value>
</property>
<property>
  <name>mapreduce.reduce.env</name>
  <value>HADOOP_MAPRED_HOME=${HADOOP_HOME}</value>
</property>
```

```
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WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License. See accompanying LICENSE file.
```

```
-->
```

```
<!-- Put site-specific property overrides in this file. -->
```

```
<configuration>
  <property>
    <name>mapreduce.framework.name</name>
    <value>yarn</value>
  </property>
  <property>
    <name>yarn.app.mapreduce.am.env</name>
    <value>HADOOP_MAPRED_HOME=${HADOOP_HOME}</value>
  </property>
  <property>
    <name>mapreduce.map.env</name>
    <value>HADOOP_MAPRED_HOME=${HADOOP_HOME}</value>
  </property>
  <property>
    <name>mapreduce.reduce.env</name>
    <value>HADOOP_MAPRED_HOME=${HADOOP_HOME}</value>
  </property>
</configuration>
```

14. Now, run the following commands on the terminal to create a directory for hadoop space, name node and data node.

```
sudo mkdir -p /usr/local/hadoop_space
sudo mkdir -p /usr/local/hadoop_space/hdfs/namenode
sudo mkdir -p /usr/local/hadoop_space/hdfs/datanode
```

Now we have successfully installed Hadoop.

15. Format the namenode as follows:

```
hdfs namenode -format
```

```
husseinfadl@husseinfadl:~$ hdfs namenode -format
2021-04-13 14:15:15,047 INFO namenode.NameNode: STARTUP_MSG:
/*****
STARTUP_MSG: Starting NameNode
STARTUP_MSG:   host = husseinfadl/127.0.1.1
STARTUP_MSG:   args = [-format]
STARTUP_MSG:   version = 3.3.0
STARTUP_MSG:   classpath = /usr/local/hadoop/etc/hadoop:/usr/local/hadoop/share/hadoop/common/lib/curator-recipes-4.2.0.jar:/usr/local/hadoop/share/hadoop/common/lib/failureaccess-1.0.jar:/usr/local/hadoop/share/hadoop/common/lib/jackson-annotations-2.10.3.jar:/usr/local/hadoop/share/hadoop/common/lib/jersey-json-1.19.jar:/usr/local/hadoop/share/hadoop/common/lib/kerb-core-1.0.1.jar:/usr/local/hadoop/share/hadoop/common/lib/jackson-core-asl-1.9.13.jar:/usr/local/hadoop/share/hadoop/common/lib/slf4j-log4j12-1.7.25.jar:/usr/local/hadoop/share/hadoop/common/lib/metrics-core-3.2.4.jar:/usr/local/hadoop/share/hadoop/common/lib/kerby-xdr-1.0.1.jar:/usr/local/hadoop/share/hadoop/common/lib/j2objc-annotations-1.1.jar:/usr/local/hadoop/share/hadoop/common/lib/javax.servlet-api-3.1.0.jar:/usr/local/hadoop/share/hadoop/common/lib/curator-client-4.2.0.jar:/usr/local/hadoop/share/hadoop/common/lib/kerb-server-1.0.1.jar:/usr/local/hadoop/share/hadoop/common/lib/zookeeper-jute-3.5.6.jar:/usr/local/hadoop/share/hadoop/common/lib/slf4j-api-1.7.25.jar:/usr/local/hadoop/share/hadoop/common/lib/curator-frame
```

This step should end by shutting down the namenode as follows:

```
2021-04-13 14:16:46,808 INFO namenode.FSImageFormatProtobuf: Image file /usr/local/hadoop_tnp/hdfs/namenode/current/fsimage.ckpt_000000000000000000 of size 406 bytes saved in 0 seconds.
2021-04-13 14:16:46,900 INFO namenode.NNStorageRetentionManager: Going to retain 1 images with txid >= 0
2021-04-13 14:16:46,909 INFO namenode.FSImage: FSImageSaver clean checkpoint: txid=0 when meet shutdown.
2021-04-13 14:16:46,909 INFO namenode.NameNode: SHUTDOWN_MSG:
/*****
SHUTDOWN_MSG: Shutting down NameNode at husseinfadl/127.0.1.1
*****/
```

16. Before starting the Hadoop Distributed File System (hdfs), we need to make sure that the rcmd type is “ssh” not “rsh” when we type the following command
- ```
pdsh -q -w localhost
```

```
husseinfadl@husseinfadl:~$ pdsh -q -w localhost
-- DSH-specific options --
Separate stderr/stdout Yes
Path prepended to cmd none
Appended to cmd none
Command: none
Full program pathname /usr/bin/pdsh
Remote program path /usr/bin/pdsh

-- Generic options --
Local username husseinfadl
Local uid 1000
Remote username husseinfadl
Rcmd type rsh
one ^C will kill pdsh No
Connect timeout (secs) 10
Command timeout (secs) 0
Fanout 32
Display hostname labels Yes
Debugging No

-- Target nodes --
localhost
```

17. If the rcmd type is “rsh” as in the above figure, type the following commands:

```
export PDSH_RCMD_TYPE=ssh
cat $HOME/.ssh/id_rsa.pub >> $HOME/.ssh/authorized_keys
chmod 0600 ~/.ssh/authorized_keys
```

Run Step 16 again to check that the rcmd type is now ssh.

If not, skip that step.

18. Start the HDFS System using the command.

```
start-dfs.sh
```

```
husseinfadl@husseinfadl:~$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [husseinfadl]
2021-04-13 14:48:34,832 WARN util.NativeCodeLoader: Unable to load nat
ive-hadoop library for your platform... using builtin-java classes whe
re applicable
```

19. Start the YARN using the command

`start-yarn.sh`

```
husseinfadl@husseinfadl:~$ start-yarn.sh
Starting resourcemanager
Starting nodemanagers
```

20. Type the following command. You should see an output similar to the one in the following figure.

`jps`

```
husseinfadl@husseinfadl:~$ jps
8484 ResourceManager
7719 NameNode
8680 NodeManager
8201 SecondaryNameNode
9517 Jps
9263 DataNode
```

Make sure these nodes are listed: (ResourceManager, NameNode, NodeManager, SecondaryNameNode, Jps and DataNode).

21. Go to localhost:9870 from the browser. You should expect the following



## Overview 'localhost:9000' (✓active)

|                |                                                            |
|----------------|------------------------------------------------------------|
| Started:       | Tue Apr 13 16:20:47 +0200 2021                             |
| Version:       | 3.3.0, raa96f1871bfd858f9bac59cf2a81ec470da649af           |
| Compiled:      | Mon Jul 06 20:44:00 +0200 2020 by brahma from branch-3.3.0 |
| Cluster ID:    | CID-31934176-56cf-44d6-aa85-3accbfae3fff                   |
| Block Pool ID: | BP-1070268464-127.0.1.1-1618323632891                      |

## Summary

Security is off.

Safemode is off.

1 files and directories, 0 blocks (0 replicated blocks, 0 erasure coded block groups) = 1 total filesystem object(s).

Heap Memory used 108.04 MB of 312 MB Heap Memory. Max Heap Memory is 3.88 GB.

